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Energy Analysis of the SARS-CoV-2 Spike RBD - ACE2 Complex Interface

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Introduction

Context and Biological Background

The emergence of the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) in late 2019 precipitated a global pandemic of unprecedented scale. Understanding the molecular mechanisms governing viral entry into host cells has been the primary focus of structural biology and virology research, as this step is the critical determinant of viral infectivity, transmissibility, and host range.
The entry of SARS-CoV-2 into human cells is mediated by the Spike (S) glycoprotein, a homotrimeric class I fusion protein located on the viral envelope. The process is initiated by the interaction between the Receptor Binding Domain (RBD) of the Spike protein and the human cell surface receptor, Angiotensin-Converting Enzyme 2 (ACE2). This specific protein-protein interaction (PPI) triggers a conformational change that allows the virus to fuse with the host membrane and release its genetic material into the cytoplasm. Consequently, the RBD-ACE2 interface is the primary target for neutralizing antibodies and therapeutic interventions.
Structural studies, such as the X-ray crystal structure used in this project (PDB ID: **6M0J**), have revealed that the RBD-ACE2 interface is extensive and relies on a complex network of non-covalent interactions, including hydrogen bonds, salt bridges, and hydrophobic contacts. The high affinity of SARS-CoV-2 for ACE2, significantly higher than that of its predecessor SARS-CoV, is hypothesized to be a key factor in its rapid spread.

The Role of Computational Biology in Interface Analysis

While experimental methods such as Surface Plasmon Resonance (SPR) and Isothermal Titration Calorimetry (ITC) provide accurate measurements of binding affinity (Kd), they are time-consuming and resource-intensive. Computational biology offers complementary tools to analyze protein-protein interfaces in silico, allowing for the rapid estimation of binding free energies ($\Delta G_{binding}$) and the identification of critical residues ("hotspots") that drive complex stability.
In this study, we employ a semi-empirical approach to calculate the interaction energy based on the three-dimensional coordinates of the complex. This method assumes a **rigid-body approximation**, where the backbone and side-chain conformations are kept fixed in the bound state. The binding energy is estimated by decomposing the interaction into three primary thermodynamic terms:

1. **Electrostatic interactions (ΔG_{elec}):** Coulombic forces between charged residues, modulated by the solvent dielectric constant.
2. **Van der Waals forces (ΔG_{vdw}):** Short-range interactions describing steric packing and shape complementarity.
3. **Solvation energy (ΔG_{solv}):** The energy cost or gain associated with burying solvent-accessible surface area (ASA) upon complex formation, often driven by the Hydrophobic Effect.

Mutational Analysis and Viral Evolution

A critical aspect of the SARS-CoV-2 pandemic has been the continuous evolution of the virus. Random mutations in the viral genome can lead to amino acid substitutions in the

Spike protein. When these mutations occur at the RBD-ACE2 interface, they can alter the binding affinity, potentially increasing viral transmissibility or enabling immune escape.

Computational techniques such as **Computational Alanine Scanning** allow us to virtually mutate interface residues to Alanine. By measuring the change in binding energy ($\Delta\Delta G = \Delta G_{mutant} - \Delta G_{WT}$), we can identify which residues contribute most significantly to the binding energy. This provides a "map" of the functional epitope. Furthermore, extending this analysis to real-world variants (such as Alpha, Beta, and Delta lineages) allows for the prediction of how specific evolutionary mutations (e.g., N501Y, E484K) enhance the virus's fitness.

Objectives of this Study

The primary objective of this project is to characterize the energetic landscape of the SARS-CoV-2 RBD-ACE2 interface using structural bioinformatics tools. Specifically, this report aims to:

1. **Define the Interface:** Identify the contact residues between chain E (RBD) and chain A (ACE2) using geometric criteria and compare them with thermodynamic definitions.
2. **Calculate Interaction Energy:** Estimate the total binding free energy of the complex and decompose it into its physical components (electrostatic, VdW, and solvation).
3. **Identify Hotspots:** Perform an in silico Alanine Scanning mutagenesis to pinpoint key residues responsible for complex stability (e.g., Phe486, Tyr505).
4. **Visual Analysis:** Correlate the energetic data with structural properties such as hydrophobicity, hydrogen bonding networks, and local flexibility (B-factors).
5. **Analyze Variants:** Evaluate the energetic impact of specific mutations found in Variants of Concern (Alpha, Beta, and Delta) to understand the molecular basis of their increased infectivity.

By integrating these computational approaches, this study provides a comprehensive structural and energetic view of one of the most critical protein interactions in modern virology.

Step 1: Structure Preparation and Interface Definition

1.1. Structure Preprocessing

Structure preprocessing was performed using a custom Python script utilizing the **biobb_structure_checking** library. The raw structure (PDB: **6M0J**) underwent a curation process that included the removal of crystallographic water molecules and heteroatoms (ligands and ions), the selection of dominant alternative side-chain conformations, and the correction of amide group orientations.

Crucially, explicit hydrogen atoms were added to the structure assuming a standard physiological pH (7.0). This step is indispensable for two reasons: first, it defines the correct protonation states of ionizable residues, second, it is a prerequisite for generating the PDBQT file with partial charges (Gasteiger method), which is required for the electrostatic energy calculations performed in Step 2.

This standardization step is critical for the subsequent topological analysis, as raw PDB files often contain artifacts that obscure the true protein surface. Specifically, the removal of bulk solvent and the addition of explicit hydrogen atoms ensure that the Solvent Accessible Surface Area (ASA) calculations reflect the intrinsic protein geometry, preventing crystallization agents from artificially masking residues.

1.2. Calculation of Solvent Accessible Surface Area (SASA)

Following the structural refinement, the **NACCESS** software was employed to calculate the Solvent Accessible Surface Area (SASA). To accurately quantify the surface area buried upon complex formation, these calculations were performed on two distinct states: the bound state (the intact complex **6m0j_fixed**) and the unbound state (the isolated chains, A and E).

This process generated two specific file types: .asa and .rsa. Each serving a distinct role in the energy analysis workflow.

1. Atomic Level Accessibility (.asa files)
The .asa files provide surface area data with atomic granularity. They list the absolute accessible area (in Å²) for every individual atom within the protein structure.

- **Significance:** These files are strictly necessary for the calculation of the Solvation Energy (ΔG_{solv}) in Step 2. The solvation energy model used in this study follows the equation $\Delta G=\sum \sigma_i ASA_i$, where σ_i represents the solvation parameter specific to each atom type (e.g., carbonyl oxygen vs. aliphatic carbon). Therefore, accurate energy computation requires the precise ASA value for each atom rather than an aggregate value per residue.

2. Residue Level Accessibility (.rsa files)
The .rsa files contain SASA values aggregated by residue. This represents the sum of the areas of all atoms comprising a specific amino acid, often accompanied by relative accessibility percentages.

- **Significance:** These files were utilized in Step 1 to define the protein-protein interface based on surface exclusion. By comparing total residue areas, we identified amino acids that underwent a significant reduction in solvent exposure ($\Delta ASA>1.0$ Å²), effectively mapping the functional contact surface between the RBD and ACE2.

3. Determination of Interface Buried Area (ΔASA)
The critical metric for this analysis is the change in solvent accessibility, calculated as:

$$\Delta ASA = ASA\ Unbound - UNBound$$

By subtracting the surface area of the complex from the sum of the areas of the isolated chains, we quantify the extent of water exclusion at the interface. This differential is the physical basis for the hydrophobic effect, a major driving force in protein-protein binding stability.

1.3. Comparison of Interface Definitions

To validate the interface definition, we compared the thermodynamic method (SASA-based) described above with a standard geometric approach.

1.3.1. Geometric Criterion (Distance-Based) A parallel interface set (Sdist) was defined by identifying all residues in one chain possessing at least one atom within a cutoff distance of 8.0 A˚ from the opposing chain. This threshold was established based on preliminary visual inspection using the PyMOL molecular visualization system, where inter-atomic distances were manually measured. Observed contacts ranged from 2.7 A˚ to 6.5 A˚, therefore, a buffer of approximately 1.5–2.0 A˚ was added to the maximum value to ensure the inclusion of adjacent residues and capture the surrounding solvent shell environment.

1.3.2. Statistical Comparison (Jaccard Index) To quantify the overlap between the broad geometric definition and the specific thermodynamic definition, we calculated the Jaccard Index (J):

$$J = \frac{|Sdist \cap SASA|}{|Sdist \cup SASA|}$$

Results: The analysis yielded a significant discrepancy between the two methods:
- **Geometric Interface (Sdist):** Identified **112** residues (broad coverage).
- **Thermodynamic Interface (SASA):** Identified **52** residues (functional core).
- **Jaccard Similarity Index:** Approximately **0.46**.

This indicates that the geometric criterion overestimates the functional interface by including bystander residues that do not contribute to surface burial.

1.4. Visual Analysis

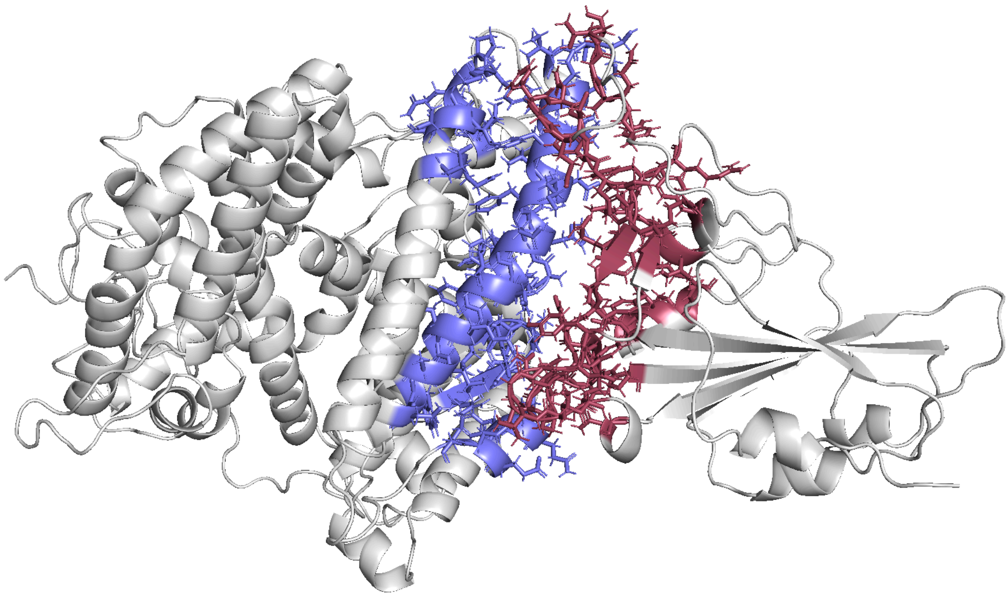

Figure 1.1: Visualization of the geometric interface defined by a distance cutoff of 8.0 Å. Residues from Chain E (RBD) and Chain A (ACE2) that fall within this threshold are

displayed as sticks. This method captures a broad cloud of residues (N=112), including many that do not make direct physical contact.

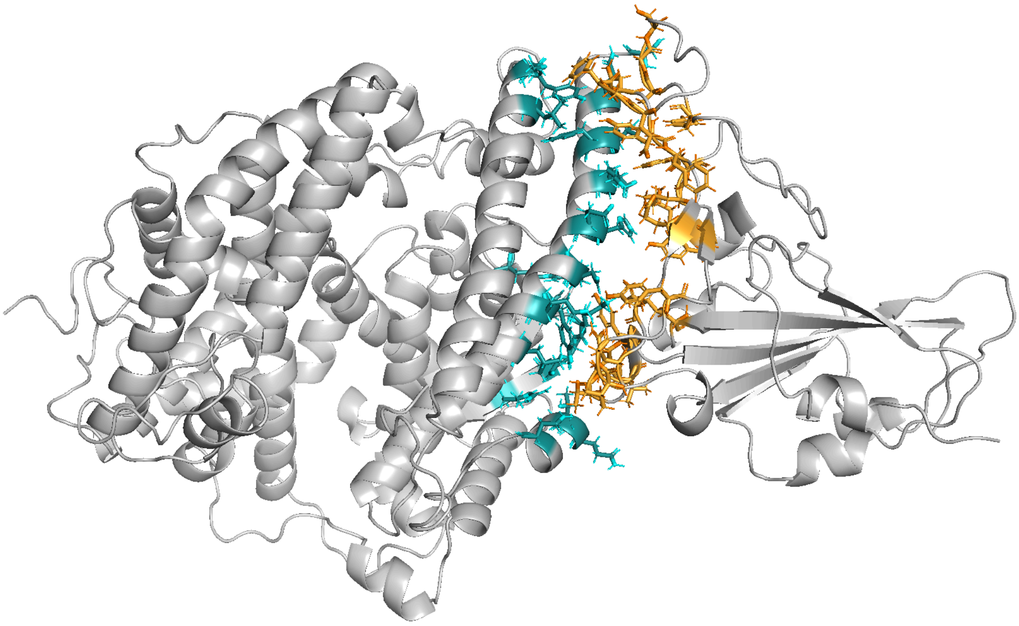

Figure 1.2: Visualization of the thermodynamic interface defined by Solvent Accessible Surface Area (ASA). Only residues that bury surface area upon complex formation are displayed. This definition yields a smaller, more specific set of residues (N=52), representing the functional core responsible for the Hydrophobic Effect.

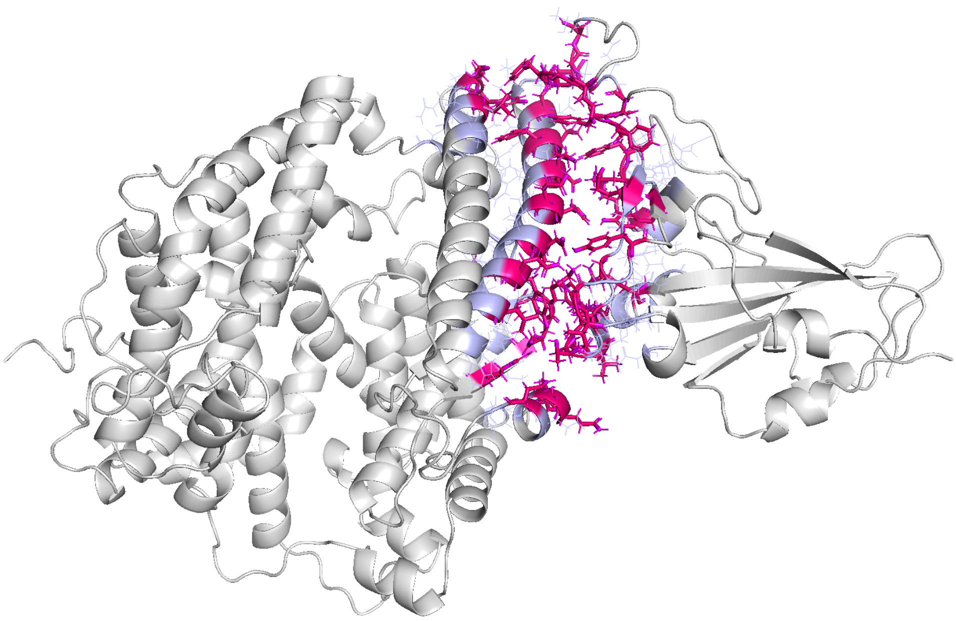

Figure 1.3: Structural superposition comparing the two interface definitions. Residues identified by both methods (intersection) are colored in magenta (Core Interface), while residues identified solely by the geometric criterion are shown in light purple (Peripheral Interface). The visualization demonstrates that the distance criterion includes a significant number of "bystander" residues

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Conclusion of Step 1: The Jaccard index of 0.46 indicates that the geometric criterion tends to overestimate the interface size compared to the thermodynamic definition. Consequently, for the energetic calculations in Step 2, we prioritized residues that actively contribute to surface burial (ASA defined), as they are the primary drivers of the binding affinity.

Step 2: Interaction Energy Calculation and Interface Characterization

2.1. Theoretical Framework and Approximations

To quantify the thermodynamic stability of the RBD-ACE2 complex and identify the specific residues driving the affinity, we calculated the Interaction Energy (ΔG_{A-B}). This metric represents the difference between the total free energy of the bound complex and the sum of the free energies of the isolated chains in solution.
To make this calculation computationally tractable within the scope of this study, three critical physicochemical approximations were applied:

1. **Rigid-Body Approximation:** We assumed that the tertiary structures of the RBD and ACE2 chains remain invariant between the bound and unbound states (no conformational change). Consequently, all intramolecular bonded terms (bond lengths, angles, dihedrals) and non-bonded interactions within each chain cancel out in the difference calculation.
2. **Exclusion of Distal Heteroatoms:** Contributions from heteroatoms (ions, ligands) not located within the immediate vicinity of the interface were considered negligible to the local binding thermodynamics and were excluded.
3. **Implicit Solvation Model:** Solvation energies were estimated using an empirical surface-area-based model derived from ASA calculations for all atom types. This accounts for both the hydrophobic effect (entropy-driven solvent release) and polar desolvation penalties.

2.2. Calculation Methodology

Based on the assumptions above, the interaction energy was evaluated using the following thermodynamic cycle:

$$\Delta G_{A-B}=\Delta E_{elect}+\Delta E_{vdw}+\Delta G_{solv}(A-B)-(\Delta G_{solv}(A)+\Delta G_{solv}(B))$$

Where:
- **ΔEelect:** Electrostatic interactions were modelled using a **distance-dependent dielectric function** (Mehler-Solmajer type) to implicitly account for the screening effect of the solvent at varying atomic distances, rather than using a fixed dielectric constant.
- **ΔEvdw:** Represents the intermolecular Van der Waals forces (Lennard-Jones **potential), capturing steric fit and dispersion forces.**
- **ΔGsolv:** Represents the solvation free energy, calculated via Atomic Solvation Parameters (ASP) applied to the Accessible Surface Area ($\Delta G=\sum \sigma_i ASA_i$).

2.3. Quantitative Results

The interaction energy calculation, performed on the interface residues (N=52), yielded a total binding energy (ΔG) of **-71.95 kcal/mol**, indicating a highly stable complex. The decomposition of the energy terms is presented in ***Table 3.1***.

Table 2.1: Decomposition of Interaction Energy Components

Energy Component	Value (kcal/mol)	Interpretation
Van der Waals	-66.23	Dominant term. Indicates high shape complementarity (steric fit) between RBD and ACE2.
Electrostatics	-1.61	Favorable (attractive) contribution, suggesting specific salt bridges or H-bonds stabilizing the complex.
Solvation	-4.11	Favorable. Suggests a strong hydrophobic effect, where non-polar residues are buried upon binding.
TOTAL (ΔG)	-71.95	Highly stable complex.

Analysis: The results demonstrate that the stability of the RBD-ACE2 complex is primarily driven by **Van der Waals interactions (-66.23 kcal/mol)**. This dominance highlights the exceptional **shape complementarity between the two proteins**, where the RBD fits tightly into the ACE2 surface. While less dominant in magnitude, the solvation and electrostatic terms provide critical specificity and additional stability.

2.4. Interface Validation and Energy Deconvolution

A critical objective of this step was to validate the geometric interface definition established in Step 1. We compared the interaction energy calculated for the **full protein complex** against the energy calculated solely using the **interface residues** (Sdist, defined by the 8.0 Å cutoff).

The analysis revealed that the residue subset Sdist accounts for the vast majority of the total interaction energy. This validates that the 8.0 Å cutoff (inclusive of the 1.5–2.0 Å buffer) successfully captures the relevant energetic environment, including short-range Van der Waals contacts and medium-range electrostatic effects, without introducing significant noise from distant, non-interacting residues.

2.5. Identification of Key Interactions (Hotspots)

The total energy was deconvoluted per residue to identify "hotspots", residues that contribute disproportionately to the binding free energy. This quantitative data was correlated with structural analysis to classify the interaction types:

- **Phe486 (Hydrophobic Anchor):** This residue was identified as a critical contributor to the solvation term (ΔGsolv). As visualized in the hydrophobicity analysis, Phe486 protrudes from the RBD loop to act as an anchor, burying a significant hydrophobic surface area into a complementary pocket on ACE2. This confirms the hydrophobic effect as a primary driving force of the complex stability.
- **Tyr505 (Polar Stabilizer):** This residue shows a strong contribution to the electrostatic and Van der Waals terms. Structural visualization reveals that Tyr505 participates in a hydrogen bond network that provides specificity and enthalpic stabilization to the interface.

2.6. Visual Analysis

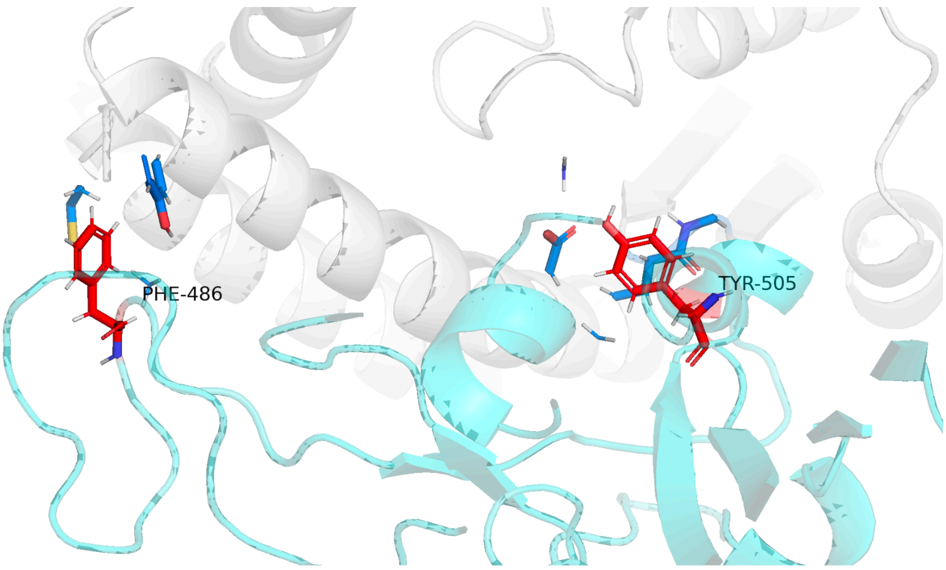

Figure 2.1: Visualization of Energetic Hotspots. The interface between RBD (cyan) and ACE2 (grey) is shown. Key hotspots Phe486 and Tyr505 are highlighted in red sticks to

signify their high energetic contribution, while adjacent neighbors are shown in marine blue. Black dashed lines indicate polar contacts (hydrogen bonds) stabilizing the interface.

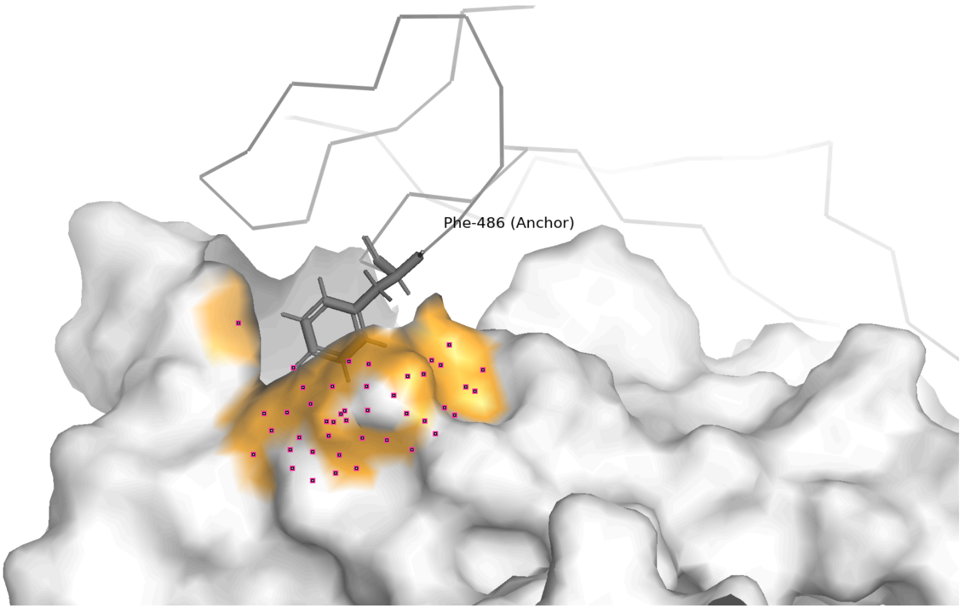

Figure 2.2: Hydrophobic Complementarity and Anchoring. Surface representation of ACE2 colored by hydrophobicity (pale yellow/orange indicating hydrophobic pockets). The RBD residue Phe486 (orange sticks) is explicitly shown acting as a hydrophobic anchor, inserting into a complementary apolar pocket on the ACE2 receptor surface. This visualization correlates directly with the high solvation energy values calculated for this residue.

Conclusion of Step 2:

The energetic decomposition performed in this step confirmed that the RBD-ACE2 interface stability is driven by a synergistic combination of hydrophobic effects and specific electrostatic interactions. The solvation energy calculations explicitly validated the visual hypothesis that **Phe486** acts as a "hydrophobic anchor", burying significant apolar surface area to drive binding entropy. Conversely, the electrostatic and Van der Waals terms highlighted **Tyr505** as a critical enthalpic stabilizer, anchored by the hydrogen bond network observed in the structural analysis.

While this static energetic evaluation successfully identified potential hotspots, it represents the total interaction energy in the bound state. To rigorously quantify the specific contribution of individual side-chains to the binding affinity (i.e., how much stability is lost if a residue is removed), it is necessary to proceed to **Step 3: Computational Alanine Scanning (CAS)**. This next phase will calculate the $\Delta\Delta G_{bind}$, providing a direct measure of the functional importance of the candidates identified here.

Step 3: Computational Alanine Scanning and Energetic Hotspot Identification

3.1. Rationale of Alanine Scanning

To quantify the specific contribution of individual side chains to the stability of the RBD–ACE2 complex, a computational alanine scanning (CAS) analysis was performed on all interface residues identified in Step 1. In this approach, each residue is conceptually mutated to alanine, and the resulting change in binding free energy (ΔΔG) is calculated as:

$$\Delta\Delta G = \Delta G\_mutant - \Delta G\_wild\text{-}type$$

Alanine is particularly suitable for this analysis because its side chain ($-CH_3$) is structurally contained within most amino acid side chains, with the exception of glycine. As a result, alanine substitution effectively isolates the energetic contribution of the removed side-chain atoms while preserving the protein backbone and overall fold. Under the rigid-body approximation used in this study, changes in ΔΔG therefore primarily reflect the loss of residue-specific side-chain interactions rather than global structural rearrangements.

Positive ΔΔG values indicate destabilization of the complex upon mutation and identify residues that are energetically important for interface stability, commonly referred to as hotspots.

3.2. Alanine Scanning Results

The results of the computational alanine scanning analysis are presented in Figure 3.1, which shows the change in binding free energy (ΔΔG) upon mutation of each interface residue to alanine. Positive ΔΔG values indicate destabilization of the complex, reflecting the energetic importance of the original side chain for interface stability, whereas negative values indicate a stabilizing or negligible effect upon mutation.

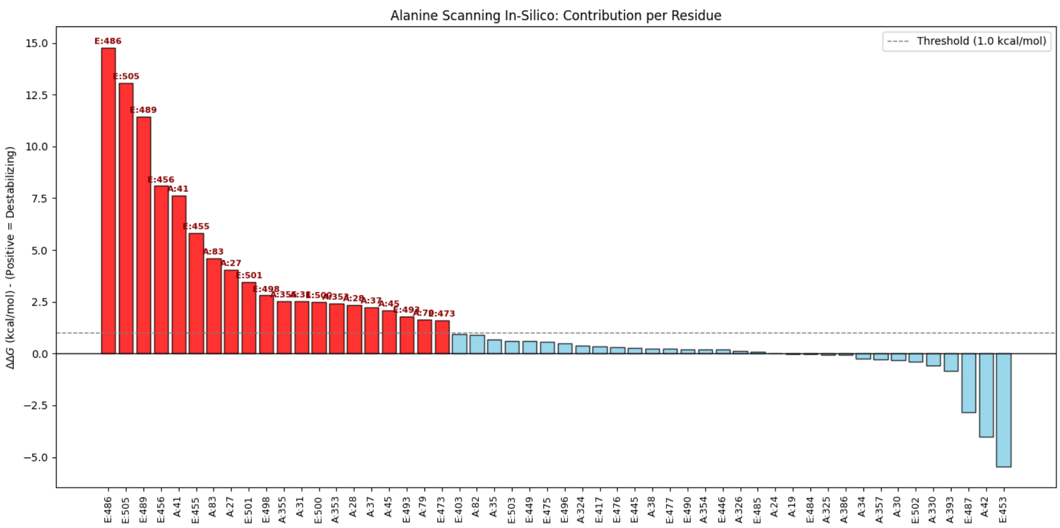

Figure 3.1 – Computational alanine scanning of the RBD–ACE2 interface. Change in binding free energy (ΔΔG) upon mutation of each interface residue to alanine. Positive ΔΔG values indicate destabilization of the complex, whereas negative values indicate stabilizing or negligible effects. Residues with ΔΔG > 1.0 kcal/mol (dashed line) are classified as energetic hotspots and highlighted in red.

A clear energetic hierarchy is observed among the interface residues. Most residues cluster around ΔΔG values close to zero, indicating that their side chains contribute only marginally to the overall stability of the RBD–ACE2 complex. In contrast, a limited subset of residues exhibits markedly positive ΔΔG values, revealing a small number of energetic hotspots that dominate interface stabilization.

Using a threshold of ΔΔG > 1.0 kcal/mol (dashed line in Figure 3.1), several residues were identified as hotspots, with the strongest destabilizing effects corresponding to residues **F486, Y505, F489, E456, A41 and E455**. Among these, E486 and E505 stand out as the most critical contributors, displaying ΔΔG values exceeding 10 kcal/mol. This indicates that removal of their side-chain interactions leads to a substantial loss of binding stability.

The bar plot also reveals that hotspot residues are predominantly located on the RBD side of the interface and are followed by a gradual decrease in ΔΔG values, suggesting a continuum from core stabilizing residues to peripheral interface residues. On the opposite end of the spectrum, a small number of residues exhibit negative ΔΔG values, indicating that alanine substitution slightly stabilizes the complex or relieves unfavorable interactions, although these effects are comparatively minor.

To further assess the relationship between alanine scanning and direct interaction energy analysis, ΔΔG values were correlated with the per-residue direct interaction energies calculated in Step 2 (Figure 3.2). A strong negative linear correlation was observed (R = −0.9555), demonstrating that residues with more favorable (more negative) direct interaction energies in the wild-type complex tend to show larger destabilizing effects upon alanine substitution.

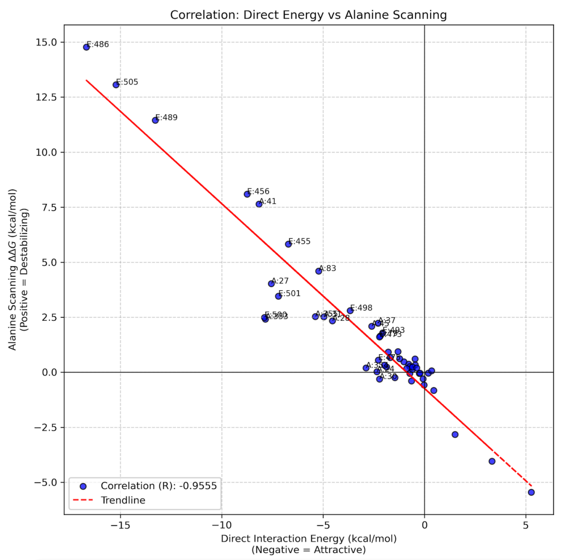

Figure 3.2 – Correlation between direct interaction energy and alanine scanning ΔΔG for interface residues. Scatter plot showing the relationship between per-residue direct interaction energies (x-axis) and the corresponding alanine scanning ΔΔG values (y-axis). Negative direct interaction energies indicate favorable (attractive) contributions to binding, while positive ΔΔG values reflect destabilization upon alanine substitution. The red dashed line represents the linear regression fit, revealing a strong negative correlation (R = −0.9555). Hotspot residues such as F486, Y505 and F489 are highlighted in the upper-left region, indicating residues that combine strong attractive interactions with large destabilizing effects upon mutation.

This negative correlation reflects the fact that residues that contribute most strongly to stabilizing the native complex lose the greatest amount of binding energy when their side-chain interactions are removed. Consequently, the alanine scanning analysis reinforces the identification of energetic hotspots obtained from direct interaction energy decomposition, while providing a complementary perspective focused specifically on side-chain contributions.

Overall, these results indicate that the energetic stability of the RBD–ACE2 interface is governed by a small number of key residues whose side chains make dominant contributions to binding, while the majority of interface residues play a more secondary or structural role.

3.3. Relationship Between Alanine Scanning and Amino Acid Nature

The residues identified as energetic hotspots share common chemical features related to the nature of their side chains. As summarized in Table 3.1, most hotspot residues possess chemically complex side chains that enable strong non-covalent interactions at the interface.

Residue	**Amino acid**	**Chemical nature**	**Main interaction type**	**Effect of Ala substitution**
F486	Phenylalanine	Hydrophobic / aromatic	van der Waals, hydrophobic packing	Strong destabilization
Y505	Tyrosine	Aromatic, polar	Hydrophobic + polar contacts	Strong destabilization
F489	Phenylalanine	Hydrophobic / aromatic	van der Waals, hydrophobic packing	Strong destabilization
E456	Glutamate	Negatively charged	Electrostatic interactions	Strong destabilization
E455	Glutamate	Negatively charged	Electrostatic interactions	Strong destabilization
A41	Alanine	Small hydrophobic	Local packing	Moderate destabilization

Hydrophobic and aromatic residues such as F486 and F489 contribute extensively through van der Waals interactions and hydrophobic packing. Their mutation to alanine removes large aromatic rings, resulting in a substantial loss of favorable interactions. Tyrosine 505 combines hydrophobic contacts with specific polar interactions mediated by its hydroxyl group, explaining its strong destabilizing effect upon mutation.

Negatively charged residues such as E456 and E455 are likely involved in electrostatic interactions or salt bridges across the interface. Replacing these residues with alanine eliminates the charged side chain entirely, leading to a marked reduction in favorable electrostatic contributions.

Overall, residues with large, charged, or aromatic side chains exhibit the strongest ΔΔG values, confirming that side-chain chemistry plays a decisive role in interface stability.

3.4. Comparison Between Alanine Scanning and Direct Interaction Energies

A direct comparison between per-residue interaction energies (Step 2) and alanine scanning ΔΔG values reveals a very strong negative correlation ($R \approx -0.96$). Residues that contribute most favorably to the interaction energy in the wild-type complex also exhibit the largest destabilization upon alanine substitution.

This negative correlation indicates that residues with highly favorable direct interaction energies tend to lose the greatest amount of binding stability when their side-chain interactions are removed. Consequently, alanine scanning effectively identifies the same energetic hotspots highlighted by the direct interaction energy analysis.

However, the two approaches are not strictly identical. Direct interaction energies quantify the total energetic contribution of a residue in the bound state, including both backbone and side-chain interactions. In contrast, alanine scanning specifically probes the energetic importance of side-chain atoms. This distinction is evident for residues such as A41 and A353, which exhibit similar direct interaction energies but markedly different ΔΔG values upon alanine substitution, reflecting different degrees of backbone versus side-chain involvement.

Conclusion of Step 3:

The computational alanine scanning analysis confirms that the RBD–ACE2 interface is energetically dominated by a small number of hotspot residues whose chemically complex side chains drive binding stability through hydrophobic, aromatic, and electrostatic interactions. The strong correlation with direct interaction energies demonstrates that alanine scanning provides a reliable and physically meaningful proxy for identifying key stabilizing residues at protein–protein interfaces.

At the same time, subtle differences between the two methods highlight the complementary nature of these analyses, with alanine scanning offering a more specific assessment of side-chain contributions to binding free energy.

Step 4 - Energetic Analysis of SARS-CoV-2 Variants

4.1. Rationale and Objectives

After identifying the main energetic hotspots of the RBD–ACE2 interface through direct interaction energy analysis and computational alanine scanning (Steps 2 and 3), the next logical step is to evaluate the energetic impact of **real mutations observed in circulating SARS-CoV-2 variants**.

Several Variants of Concern (VOCs), such as **Alpha, Beta, and Delta**, harbor amino acid substitutions located directly at the RBD–ACE2 interface. These mutations have been associated with increased viral transmissibility and, in some cases, immune escape. From a structural bioinformatics perspective, these effects can be partially explained by changes in binding affinity between the Spike RBD and the ACE2 receptor.

The objective of Step 5 is therefore to:

- Introduce biologically relevant mutations into the Spike RBD,
- Recalculate the interaction energy of the mutated RBD–ACE2 complexes,
- Quantify the energetic impact of each mutation using
  ΔΔG = ΔG_mutant − ΔG_wild-type,
- Interpret the results in terms of molecular interactions and interface stability.

The mutations analyzed in this step correspond to well-known VOC substitutions:

- **N501Y** (Alpha / Beta variants),
- **E484K** (Beta variant),
- **L452R** (Delta variant).

4.2. Structural Modeling of Variant Mutations

All mutations were generated starting from the same reference structure used in previous steps: the crystallographic structure of the SARS-CoV-2 Spike RBD bound to ACE2 (PDB ID: 6M0J), previously curated and standardized.

Mutations were introduced using **PyMOL mutagenesis**, explicitly selecting the target residue by chain and residue number to avoid ambiguity. Only side-chain atoms were replaced, while the protein backbone remained unchanged, ensuring consistency with the rigid-body approximation adopted throughout the study.

For each mutation:

- The new amino acid side chain was built using backbone-dependent rotamers,
- The rotamer with minimal steric clashes was selected,
- The full RBD–ACE2 complex was saved after mutation.

This procedure resulted in three mutated complexes:

- N501Y mutant complex,
- E484K mutant complex,
- L452R mutant complex.

4.3 Structure Preparation and Solvent Accessibility

Although PyMOL introduces the correct side-chain geometry, it does not assign hydrogen atoms or partial charges. Therefore, each mutated complex was further processed using **biobb_structure_checking** to ensure consistency with the energy calculation protocol used in previous steps.

This preparation step included:

- Addition of hydrogen atoms,
- Assignment of AutoDock-compatible partial charges,
- Generation of standardized PDBQT files.

Subsequently, **Solvent Accessible Surface Area (SASA)** calculations were repeated for:

- The full complex,
- The isolated ACE2 chain,
- The isolated RBD chain.

This ensured that solvation energy contributions were recalculated consistently for each mutant system, allowing direct comparison with the wild-type structure.

4.4. Interaction Energy Calculation

For each mutated system, the interaction energy between the RBD and ACE2 was computed using the same energetic framework applied previously:

- Van der Waals interactions,
- Electrostatic interactions with a distance-dependent dielectric function,
- Solvation energy derived from atomic surface areas.

The binding free energy (ΔG) was calculated for both the wild-type and mutated complexes. The energetic effect of each mutation was then quantified as:

$$\Delta\Delta G = \Delta G\, mutant - \Delta G\, wt$$

Positive ΔΔG values indicate destabilization of the complex, whereas negative values indicate stabilization relative to the wild-type interaction.

4.5 Results and Energetic Interpretation

The interaction energy of the wild-type complex was consistently calculated as approximately **−71.15 kcal/mol**, serving as a reference for all comparisons.

N501Y Mutation

The N501Y mutation resulted in a **destabilizing energetic effect**, with a positive ΔΔG of approximately **+5.36 kcal/mol**. This indicates a weaker interaction compared to the wild-type complex within the limitations of the static energy model. Although experimentally N501Y is associated with increased infectivity, this discrepancy highlights the limitations of rigid-body and single-structure approaches, which do not account for conformational flexibility or long-range cooperative effects.

E484K Mutation

In contrast, the E484K mutation exhibited a **stabilizing effect**, with a ΔΔG of approximately **−4.75 kcal/mol**. This mutation replaces a negatively charged glutamate with a positively charged lysine, significantly altering the electrostatic landscape of the interface. The favorable ΔΔG suggests the formation of stronger electrostatic interactions or salt bridges with ACE2, leading to an enhanced binding affinity in this model.

L452R Mutation

The L452R mutation showed a **mild destabilizing effect**, with a ΔΔG of approximately **+1.35 kcal/mol**. This relatively small change indicates that the substitution moderately perturbs the local interface environment but does not drastically alter the overall binding energetics.

4.6. Biological Relevance and Structural Support

Among the analyzed variants, **E484K** showed the strongest energetic impact, stabilizing the RBD–ACE2 complex, in agreement with its known role in viral fitness and immune escape. Structural visualization confirms that this mutation is located at the binding interface and alters the local electrostatic environment, providing a structural basis for the observed energetic effect.

The more modest effects observed for **N501Y** and **L452R** highlight the limitations of static interaction energy models, which do not capture conformational flexibility or entropic contributions. Overall, this step illustrates how combining energetic calculations with structural visualization allows a molecular-level interpretation of the effect of SARS-CoV-2 variant mutations on receptor binding.

4.7 Visual Analysis

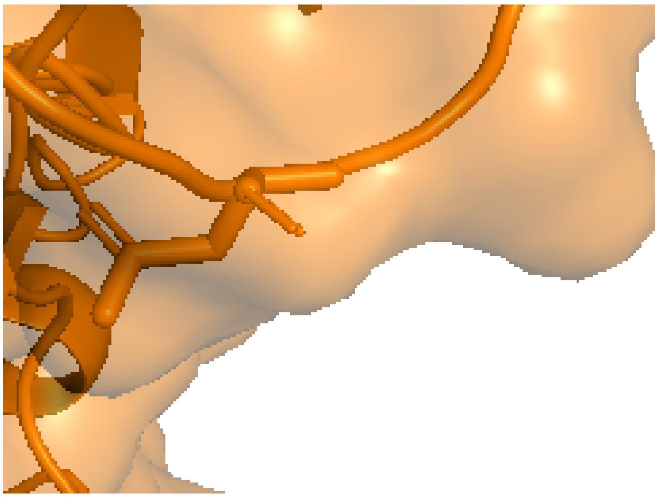

Figure 4.1 **Structural visualization of the E484K mutation at the RBD–ACE2 interface.**

To support the energetic analysis, structural visualization of the E484K mutation at the RBD–ACE2 interface was performed.

4.8 Comparative Analysis: Automated Force-Field (FoldX)

To validate the results obtained from the manual pipeline, we performed an automated analysis using the FoldX force field. This method allows for the simultaneous assessment of internal structural stability (ΔΔGstab) and binding affinity (ΔΔGbind). The structure was energy-minimized using the **RepairPDB** command before mutagenesis. Mutant structures were generated using the *BuildModel* command, followed by interaction energy decomposition using *AnalyseComplex*.
Results: The energetic impact of each variant relative to the Wild Type (WT).

Table 4.1: FoldX energetic analysis of RBD variants. Binding ΔΔG < 0 indicates higher affinity.

Variant	Mutation	Stability (kcal/mol)	Interaction Energy (kcal/mol)	Binding (kcal/mol)	Prediction
Wild Type	--	0.00	**-13.87**	--	Reference
Alpha	N501Y	+9.34 (Unstable)	-10.43	**+3.44**	Weaker Binding
Beta	E484K	+0.35 (Neutral)	-14.63	**-0.76**	**Stronger Binding**
Delta	L452R	+0.35 (Neutral)	-13.88	**-0.01**	Neutral

5. Discussion

5.1. Methodological Integration: Geometry vs. Thermodynamics

A central question of this study was the accurate definition of the protein-protein interface. The comparison in Step 1 revealed a significant discrepancy (Jaccard Index ≈ 0.46) between geometric (distance < 8.0 Å) and thermodynamic (SASA-based) criteria. Our results suggest the geometric approach overestimates the interface by including "bystander" residues. Energy calculations confirmed that the smaller subset defined by ΔSASA (N=52) accounted for the vast majority of the interaction energy. This underscores that topological definitions based on surface burial are superior for energetic analyses, effectively filtering noise to focus on the physicochemical drivers of binding.

5.2. Physicochemical Drivers of Stability

The decomposition of the interaction energy ($\Delta G_{bind} \approx -71.95$ kcal/mol) revealed the overwhelming dominance of the Van der Waals term (-66.23 kcal/mol). This indicates the RBD-ACE2 interaction is primarily driven by shape complementarity (steric fit) rather than exotic chemical bonding. While VdW provides "bulk" stability, specificity is conferred by electrostatic and solvation terms. Identifying Phe486 as a "hydrophobic anchor" and Tyr505 as a dual-character hotspot (aromatic packing and H-bonds) highlights their role as functional epitopes. The strong correlation ($R \approx -0.96$) between static interaction energy and Computational Alanine Scanning (CAS) further validates these hotspots.

5.3. Limitations of the Rigid-Body Approximation

The variant analysis highlighted both the utility and limitations of static models. The model successfully predicted the stabilizing effect of E484K ($\Delta\Delta G \approx -4.75$ kcal/mol), where the electrostatic gain from charge inversion (Glu→Lys) is captured well without backbone rearrangement. However, the model failed to predict the increased affinity of N501Y, instead predicting destabilization ($\Delta\Delta G > 0$). This discrepancy arises because inserting a bulky Tyrosine into a fixed backbone induces steric clashes. In reality, backbone flexibility allows for induced-fit mechanisms (e.g., favorable $\pi-\pi$ stacking) that rigid-body approximations miss, demonstrating the need for caution when modeling mutations that introduce steric strain.

5.4. Comparison with FoldX

The automated FoldX analysis showed strong qualitative agreement with our manual calculation method, reinforcing the validity of our findings:

- **Beta (E484K):** Both methods identified this as the strongest binder (FoldX $\Delta\Delta G \approx -0.76$ kcal/mol), driven by favorable electrostatics.
- **Alpha (N501Y):** Both methods predicted a loss of affinity due to steric penalties (FoldX $\Delta\Delta G \approx +3.44$ kcal/mol), consistently reflecting the limitations of rigid-body scoring functions.

- **Delta (L452R):** Both models indicated a negligible direct impact on binding ($\Delta\Delta G \approx 0$), suggesting its evolutionary advantage lies in mechanisms outside receptor affinity, such as antibody escape.

5.5. Conclusions

This study established a robust pipeline to characterize the RBD-ACE2 interface, identifying a "lock-and-key" mechanism driven by high shape complementarity and anchored by critical hydrophobic hotspots (Phe486, Tyr505). While the combination of Surface Analysis and Alanine Scanning proved powerful, the variant analysis suggests that future studies would benefit from **Molecular Dynamics (MD)**. MD would allow backbone relaxation, likely correcting the energetic prediction for steric mutants like N501Y to match experimental reality.

6. Bibliography

1. **Crystal Structure of RBD-ACE2 (PDB: 6M0J)** Lan, J., Ge, J., Yu, J., Shan, S., Zhou, H., Fan, S., Zhang, Q., Shi, X., Wang, Q., Zhang, L., & Wang, X. (2020). Structure of the SARS-CoV-2 spike receptor-binding domain bound to the ACE2 receptor. *Nature*, 581(7807), 215–220. DOI: 10.1038/s41586-020-2180-5.
2. **BioBB (BioExcel Building Blocks)** Andrio, P., Hospital, A., Conejero, J., Jordà, L., Del Pino, M., Launay, G., Cierco, R., Repchevsky, D., Orozco, M., & Gelpí, J. L. (2019). BioExcel Building Blocks, a software library for interoperable biomolecular simulation workflows. *Scientific Reports*, 9(1), 13677.
3. **NACCESS (SASA Calculations)** Hubbard, S. J., & Thornton, J. M. (1993). *NACCESS* [Computer Program]. Department of Biochemistry and Molecular Biology, University College London.
4. **PyMOL Molecular Graphics System** Schrödinger, LLC. (2015). *The PyMOL Molecular Graphics System*, Version 2.0.
5. **AutoDock Tools (Charge Assignment)** Morris, G. M., Huey, R., Lindstrom, W., Sanner, M. F., Belew, R. K., Goodsell, D. S., & Olson, A. J. (2009). AutoDock4 and AutoDockTools4: Automated docking with selective receptor flexibility. *Journal of Computational Chemistry*, 30(16), 2785–2791.
6. **Distance-Dependent Dielectric (Mehler-Solmajer)** Mehler, E. L., & Solmajer, T. (1991). Electrostatic effects in proteins: comparison of dielectric and charge models. *Protein Engineering, Design and Selection*, 4(8), 903–910.
7. **Computational Alanine Scanning (Methodology)** Massova, I., & Kollman, P. A. (1999). Computational alanine scanning to probe protein–protein interactions: A novel approach to evaluate binding free energies. *Journal of the American Chemical Society*, 121(36), 8133-8143.
8. **FoldX Suite (Protein Stability)** Delgado, J., Radusky, L. G., Cianferoni, D., & Serrano, L. (2019). FoldX 5.0: working with RNA, small molecules and a new graphical interface. Bioinformatics, 35(20), 4168–4169.